

Date: 04 JAN 2025**Regd. No.**

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Sub Code:23CS/IT/AD/AM2T04**MIC23**

B. Tech II Semester Supplementary Examinations, Dec-2024

DATA STRUCTURES

(Common to CSE, IT, AI&DS, AI&ML)

Time: 3 hours**Max Marks: 70M**

- Note:** 1. Question paper consists of two parts (**Part A and Part B**)
 2. Answer all sub questions from **Part A**
 3. Answer all questions from **Part B** with **either or choice**

PART - A			10 X 2 = 20 M		
			Marks	BTL	CO
I.	a)	Differentiate Linear and Non-Linear Data Structures	2 M	L3	CO1
	b)	Define Time and Space Complexity.	2 M	L2	CO1
	c)	Write the advantages of double linked list over single linked list.	2 M	L2	CO2
	d)	Write the advantage of Circular Linked List.	2 M	L2	CO2
	e)	List any two applications of stack.	2 M	L3	CO3
	f)	Write the equivalent postfix expression for a + b * c	2 M	L3	CO3
	g)	Define the operations of Queue.	2 M	L2	CO4
	h)	Define a Deque.	2 M	L2	CO4
	i)	What is a Hash table.	2 M	L2	CO5
	j)	What is the role of Hash Function in hashing.	2 M	L3	CO5
PART - B			5 X 10 = 50 M		
UNIT - I					
1.	a)	Explain about time and space complexities	5M	L2	CO1
	b)	Write a program that uses a function to perform Linear Search.	5M	L3	CO1
(OR)					
2.	a)	Apply Bubble Sort on the following list of elements. 35, 22, 49, 85, 14, 61, 76.	5M	L3	CO1
	b)	Write a program that uses a function to perform selection sort.	5M	L3	CO1
UNIT - II					
3.	a)	Explain about insertion operations on Single Linked List	5M	L2	CO2
	b)	Write the function(s) for insertion operations on Double Linked List	5M	L3	CO2
(OR)					
4.	a)	Differentiate Arrays and Linked Lists	5M	L3	CO2
	b)	Explain about representation of Sparse Matrix using linked list.	5M	L2	CO2
UNIT - III					
5.	a)	Write a program to implement stack using array.	5M	L3	CO3
	b)	Explain how the operations performed on stack using array.	5M	L2	CO3
(OR)					
6.	a)	Convert the following expression from infix to postfix using stack. (a + b) * (c - d) / e	5M	L3	CO3

	b)	Explain about postfix evaluation using stack with an example.	5M	L2	CO3
UNIT – IV					
7.	a)	Explain how the operations performed on queue using array.	5M	L2	CO4
	b)	Write a program to implement queue using array.	5M	L3	CO4
(OR)					
8.	a)	Explain about applications of Queue.	5M	L3	CO4
	b)	Explain about operations performed on Deque.	5M	L2	CO4
UNIT – V					
9.	a)	Explain about Binary Tree representations with examples.	5M	L2	CO5
	b)	Explain all the cases of deleting an element from Binary Search Tree with examples.	5M	L3	CO5
(OR)					
10.	a)	Define Hashing. Explain about Separate Chaining with an example.	5M	L2	CO5
	b)	Explain about the application of hashing: Caching	5M	L2	CO5

***** End of the Question Paper *****